10.1: Photosynthesis feeds the biosphere

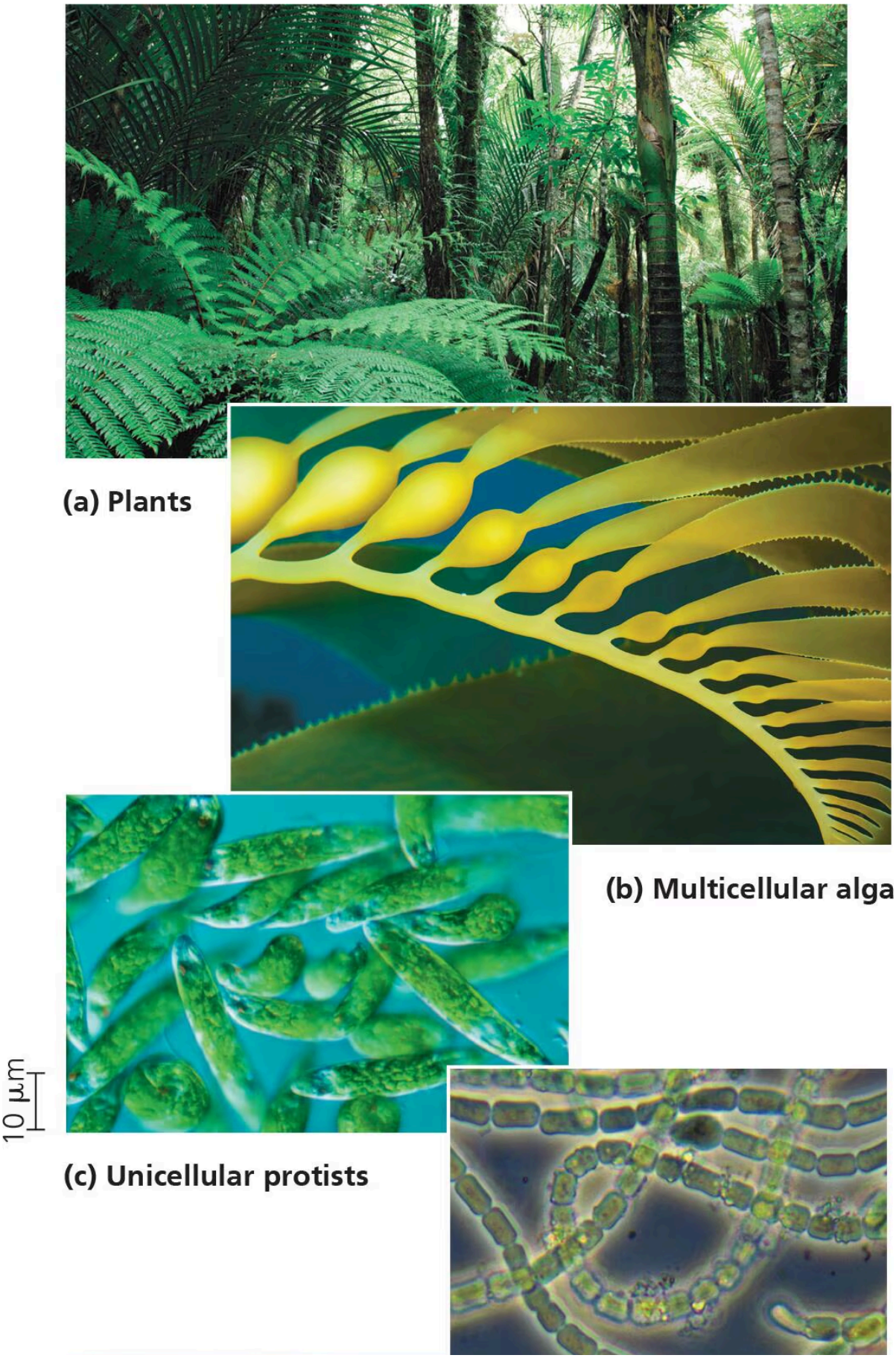
The conversion process that transforms the energy of sunlight into chemical energy stored in sugars and other organic molecules is called **photosynthesis**. Let's begin by placing photosynthesis in its ecological context.

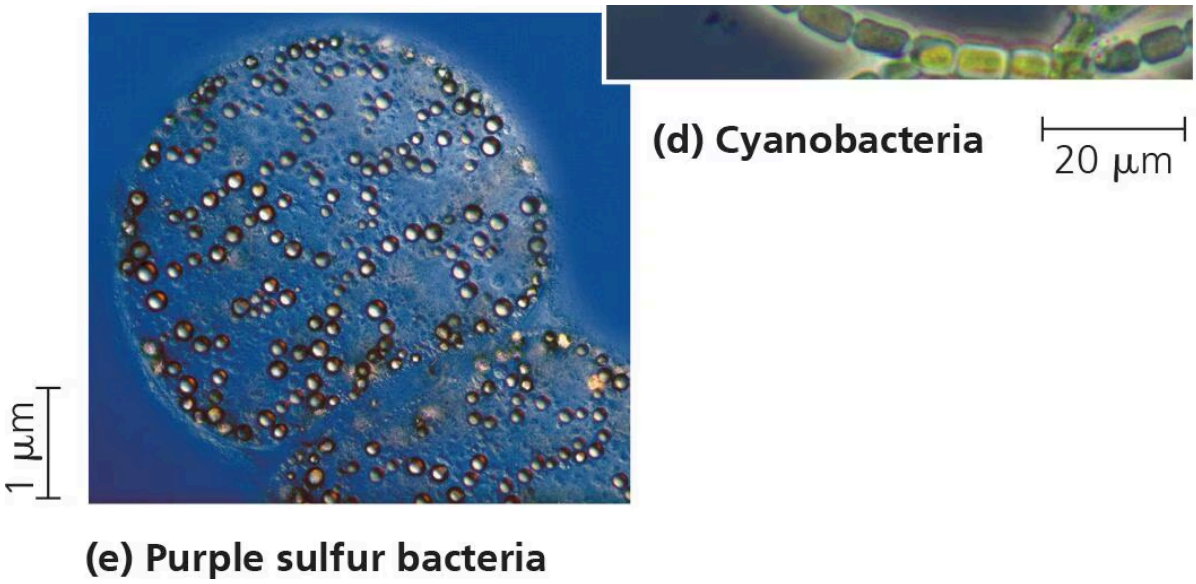
Photosynthesis nourishes almost the entire living world directly or indirectly. An organism acquires the organic compounds it uses for energy and carbon skeletons by one of two major modes: autotrophic nutrition or heterotrophic nutrition. **Autotrophs** are "self-feeders" (*auto-* means "self," and *trophos* means "feeder"); they sustain themselves without eating anything derived from other living beings. Autotrophs produce their organic molecules from CO_2 and other inorganic raw materials obtained from the environment. They are the ultimate sources of organic compounds for all nonautotrophic organisms, and for this reason, biologists refer to autotrophs as the *producers* of the biosphere.

Almost all plants are autotrophs; the only nutrients they require are water and minerals from the soil and CO_2 from the air. Specifically, plants are *photo*autotrophs, organisms that use light as a source of energy to synthesize organic substances **(Figure 10.1)**. Photosynthesis also occurs in algae, certain other protists, and some prokaryotes **(Figure 10.2)**. In this chapter, we will touch on these other groups in passing, but our emphasis will be on plants. Variations in autotrophic nutrition that occur in prokaryotes and algae will be described in Concept 27.3.

Figure 10.2

Photoautotrophs.


(a) Plants
(b) Multicellular alga
10 µm
(c) Unicellular protists



These organisms use light energy to drive the synthesis of organic molecules from carbon dioxide and (in most cases) water. They feed themselves and the entire living world. On land, ***(a)*** *plants are the predominant producers of food. In aquatic environments, photoautotrophs include unicellular and* ***(b)*** *multicellular algae, such as this kelp;* ***(c)*** *some non-algal unicellular protists, such as Euglena;* ***(d)*** *the prokaryotes called cyanobacteria; and* ***(e)*** *other photosynthetic prokaryotes, such as these purple sulfur bacteria, which produce sulfur (the yellow globules within the cells) (c–e, LMs).*

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The Flow of Carbon Atoms in Producers, Consumers, and Decomposers

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Heterotrophs are unable to make their own food; they live on compounds produced by other organisms (*hetero-* means “other”). Heterotrophs are the biosphere’s *consumers*. The most obvious “other-feeding” occurs when an animal eats plants or other organisms, but heterotrophic nutrition may be more subtle. Some heterotrophs consume the remains of other organisms and organic litter such as feces, and fallen leaves; these types of heterotrophs are known as *decomposers*. Most fungi and many types of prokaryotes get their nourishment this way. Earth’s supply of fossil fuels was formed from remains of organisms that died hundreds of millions of years ago. In a sense, then, fossil fuels represent stores of the sun’s energy from the distant past. Almost all heterotrophs, including humans, are completely dependent, either directly or indirectly, on photoautotrophs for food—and also for oxygen, a by-product of photosynthesis.